

Anna Nariyeva

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Portfolio

EDUCATION

Georgia Institute of Technology

B.S. in Mechanical Engineering, Concentration in Manufacturing — GPA: 3.50/4.00

May. 2026 – Exp. May. 2028

Atlanta, GA

The Pennsylvania State University

B.S. in Mechanical Engineering, Minor in Mechatronics — GPA: 3.70/4.00

Jan. 2024 – May. 2026

Abington, PA

Yonsei University — Korean Language Institute

Intensive Korean Course - Intermediate Level

Aug. 2022 – Nov. 2023

Seoul, South Korea

ENGINEERING RESEARCH

Part-time Undergraduate Biomechanical Research Assistant

Jan. 2025 – May. 2026

The Pennsylvania State University

Abington, PA

- Performed mechanical testing and data analysis of 100+ 3D-printed polypropylene orthotic braces using Tinius Olsen (tensile testing) equipment to quantify tensile, compressive, and torsional performance parameters on behalf of Korthotics, an Australian orthotics company.
- Analyzed experimental data to investigate the effects of print orientation, material type, extrusion temperature, and layer height on the strength and flexibility of 3D-printed components for ankle-foot orthoses (AFOs).
- Applied finite element analysis (FEA) in Fusion 360 and compared simulated and experimental results, calibrated material models, and improved the accuracy of digital simulations of additively-manufactured components.
- Presented findings at the ASB East Regional Conference at the University Delaware and collaborated with the biomechanical department to develop production methodologies that minimize the cost of manufacturing.

Full-time Multi-campus REU - Biomechanical Research Assistant

May. 2025 – Aug. 2025

The Pennsylvania State University

University Park and Abington, PA

- Quantified the flexural strength and stiffness of 20+ conventionally and commercially fabricated 3D-printed ankle-foot orthoses (AFOs) to establish performance benchmarks for clinically used devices.
- Eliminated testing inaccuracy by designing testing protocols and custom fixtures compatible with Tinius Olsen equipment to measure strength and deformation.
- Performed finite element analysis (FEA) to evaluate AFO designs and investigate design solutions that meet specified strength and performance requirements.
- Worked closely with Dr. Mukul Talaty through weekly research meetings to interpret experimental findings, refine testing methodologies, and develop independent problem-solving skills.

EXPERIENCE

Part-time Engineering Laboratory Assistant

Sep. 2025 – May. 2026

The Pennsylvania State University

Abington, PA

- Assembled, calibrated, and maintained 10+ 3D printers and mechanical testing equipment to support undergraduate engineering coursework and laboratory experiments.
- Trained and supervised 30+ undergraduate students in the safe operation of 3D printers and mechanical testing equipment, promoting laboratory safety and reliable experimental procedures.
- Troubleshooted laboratory hardware and engineering software, including Prusa and Raise3D printers, SolidWorks, and Fusion 360, while assisting with experimental documentation.

CONFERENCES

ASB East Regional Conference, University of Delaware

April, 2026

Mechanical Performance of Polypropylene for FDM-Based 3D-Printed Ankle-Foot Orthoses

Multi-Campus REU, Penn State University Park

July, 2025

Comparing the Mechanical Properties for Conventionally Fabricated and 3D-Printed Ankle-Foot Orthoses

AWARDS

Multi-campus REU Summer Program Scholarship

SKILLS

Engineering: CAD, Finite Element Analysis (FEA), Mechanical Testing, Additive Manufacturing, C++

Software: SolidWorks, Fusion 360, MATLAB, NI Multisim, Quartus Prime, Tinius Olsen Horizon

Manufacturing & Prototyping: FDM printing, SLS printing, FGF printing, PrusaSlicer, Cura

Languages: English (Fluent), Spanish (Native), Russian (Native), Korean (Intermediate)